7. Worksheet: Diversity Synthesis

El Park; Z620: Quantitative Biodiversity, Indiana University

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OVERVIEW

In this worksheet, you will conduct exercises that reinforce fundamental concepts of biodiversity. First, you will construct a site-by-species matrix by sampling confectionery taxa from a source community. Second, you will make a preference-profile matrix, reflecting each student's favorite confectionery taxa. With this primary data structure, you will then answer questions and generate figures using tools from previous weeks, along with wrangling techniques that we learned about in class.

Directions:

1. In the Markdown version of this document in your cloned repo, change "Student Name" on line 3 (above) to your name.
2. Complete as much of the worksheet as possible during class.
3. Refer to previous handouts to help with developing of questions and writing of code.
4. Answer questions in the worksheet. Space for your answer is provided in this document and indicated by the ">" character. If you need a second paragraph be sure to start the first line with ">". You should notice that the answer is highlighted in green by RStudio (color may vary if you changed the editor theme).
5. Before you leave the classroom, **push** this file to your GitHub repo.
6. For the assignment portion of the worksheet, follow the directions at the bottom of this file.
7. When you are done, **Knit** the text and code into a PDF file.
8. After Knitting, submit the completed exercise by creating a **pull request** via GitHub. Your pull request should include this file `7.DiversitySynthesis_Worskheet.Rmd` and the PDF output of `Knitr` (`DiversitySynthesis_Worskheet.pdf`).

QUANTITATIVE CONFECTIONOLOGY

We will construct a site-by-species matrix using confectionery taxa (i.e, jelly beans). The instructors have created a **source community** with known abundance (N) and richness (S). Like a real biological community, the species abundances are unevenly distributed such that a few jelly bean types are common while most are rare. Each student will sample the source community and bin their jelly beans into operational taxonomic units (OTUs).

SAMPLING PROTOCOL: SITE-BY-SPECIES MATRIX

1. From the well-mixed source community, each student should take one Dixie Cup full of individuals.

2. At your desk, sort the jelly beans into different types (i.e., OTUs), and quantify the abundance of each OTU.

3. Working with other students, merge data into a site-by-species matrix with dimensions equal to the number of students (rows) and taxa (columns)

4. Create a worksheet (e.g., Google sheet) and share the site-by-species matrix with the class.

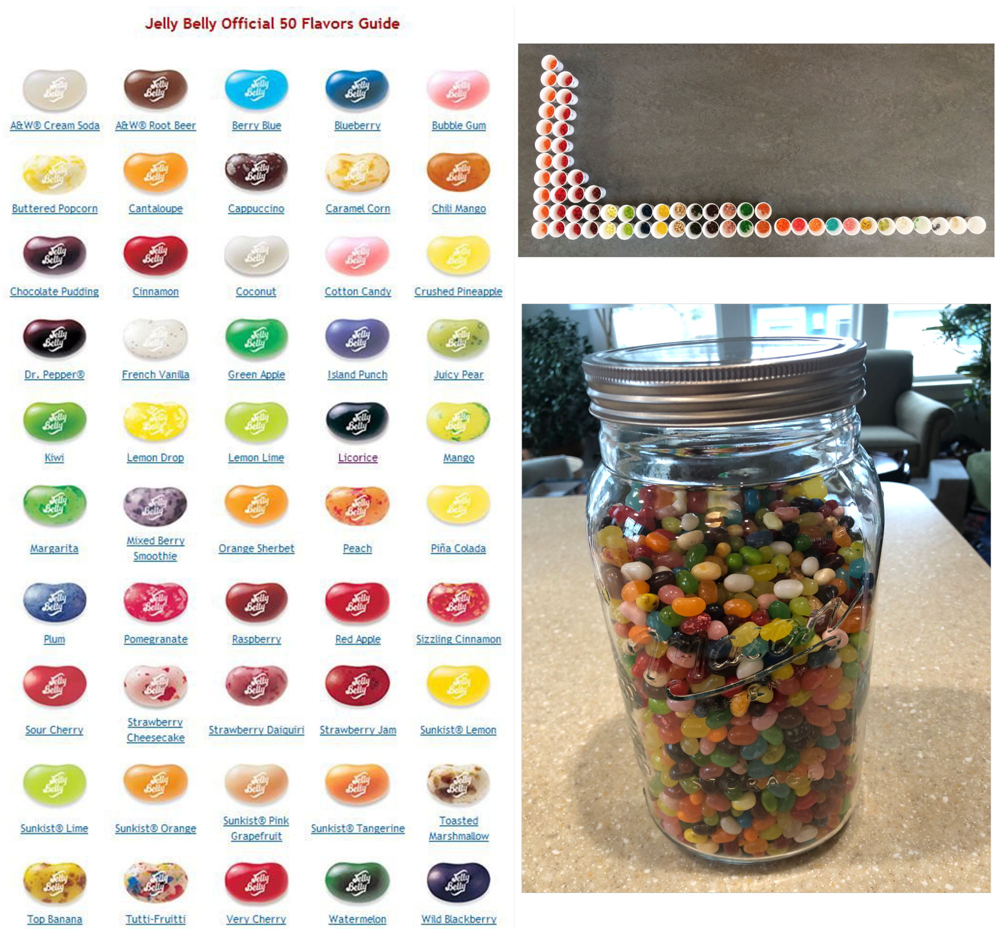


Figure 1: **Left**: taxonomic key, **Top right**: rank abundance distribution, **Bottom right**: source community

SAMPLING PROTOCOL: PREFERENCE-PROFILE MATRIX

1. With your individual sample only, each student should choose their top 5-10 preferred taxa based on flavor, color, sheen, etc.
2. Working with other students, merge data into preference-profile incidence matrix where 1 = preferred and 0 = non-preferred taxa.
3. Create a worksheet (e.g., Google sheet) and share the preference-profile matrix with the class.

1) R SETUP

In the R code chunk below, please provide the code to: 1) Clear your R environment, 2) Print your current working directory, 3) Set your working directory to your `Week5-Confection/` folder, and 4) Load the `vegan` R package (be sure to install first if you have not already).

```
rm(list = ls())
getwd()
```

```
## [1] "/cloud/project/QB2025_Park/Week5-Confection"
```

```
setwd("/cloud/project/QB2025_Park/Week5-Confection")
package.list <- c('vegan', 'ade4', 'viridis', 'gplots', 'BiodiversityR', 'indicspecies')
for (package in package.list) {
  if (!require(package, character.only = TRUE, quietly = TRUE)) {
    install.packages(package)
    library(package, character.only = TRUE)
  }
}
```

```
##
## Attaching package: 'gplots'

## The following object is masked from 'package:stats':
##
##     lowess

## Warning in fun(libname, pkgname): couldn't connect to display ":0"

## BiodiversityR 2.17-1.1: Use command BiodiversityRGUI() to launch the Graphical User Interface;
## to see changes use BiodiversityRGUI(changeLog=TRUE, backward.compatibility.messages=TRUE)
```

```
install.packages("Rcmdr", dependencies = TRUE)
```

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.4'
## (as 'lib' is unspecified)
```

DATA ANALYSIS

Question 1: In the space below, generate a rarefaction plot for all samples of the source community. Based on these results, discuss how individual vs. collective sampling efforts capture the diversity of the source community.

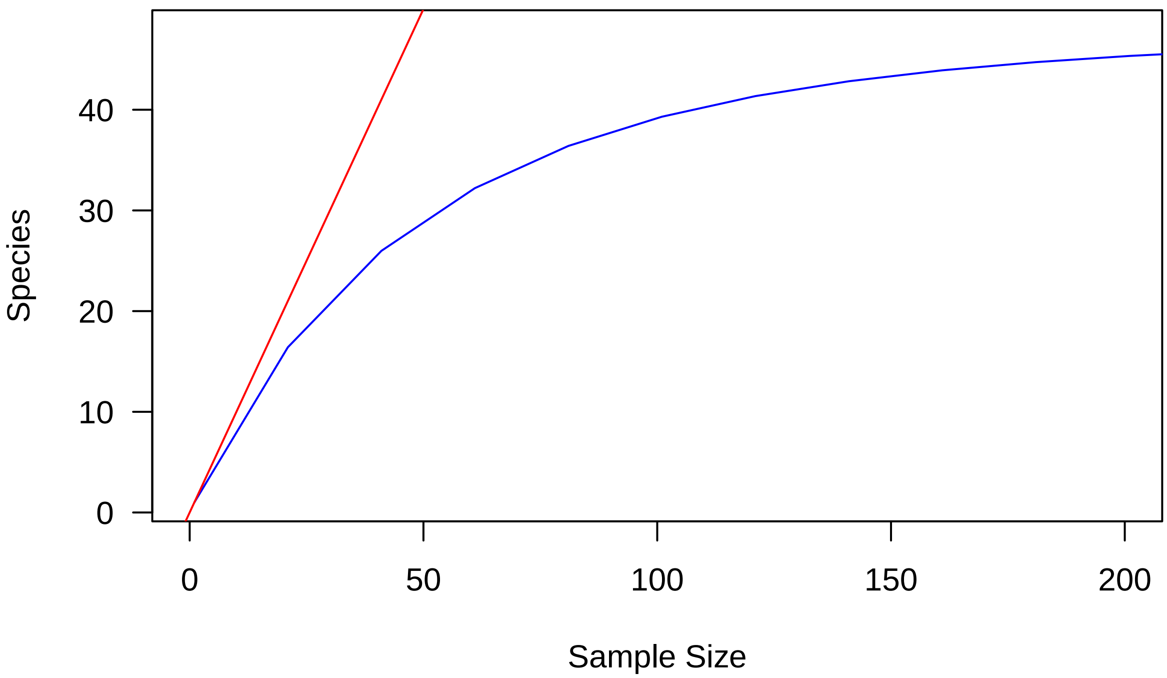
```
s.obs <- function(x=""){
rowSums(x>0) * 1
}
scjelly<-read.csv("data/jelly.source.comm.csv", sep =",", header = TRUE, row.names=1)
scjelly.t<-as.data.frame(t(scjelly))
min<-min(rowSums(scjelly.t))
s.rarefy<-rarefy(x=scjelly.t, sample=min, se=TRUE)
```

```
## Warning in rarefy(x = scjelly.t, sample = min, se = TRUE): most observed count
## data have counts 1, but smallest count is 13
```

```r
plot.new()
rarecurve(x=scjelly.t, step=20, col="blue",cex=0.6, las=1, xlim = c(0, 200))
```

```
## Warning in rarecurve(x = scjelly.t, step = 20, col = "blue", cex = 0.6, : most
## observed count data have counts 1, but smallest count is 13
```

```r
abline(0, 1, col='red')
text(1500, 1500, "1:1", pos=2, col='red')
```

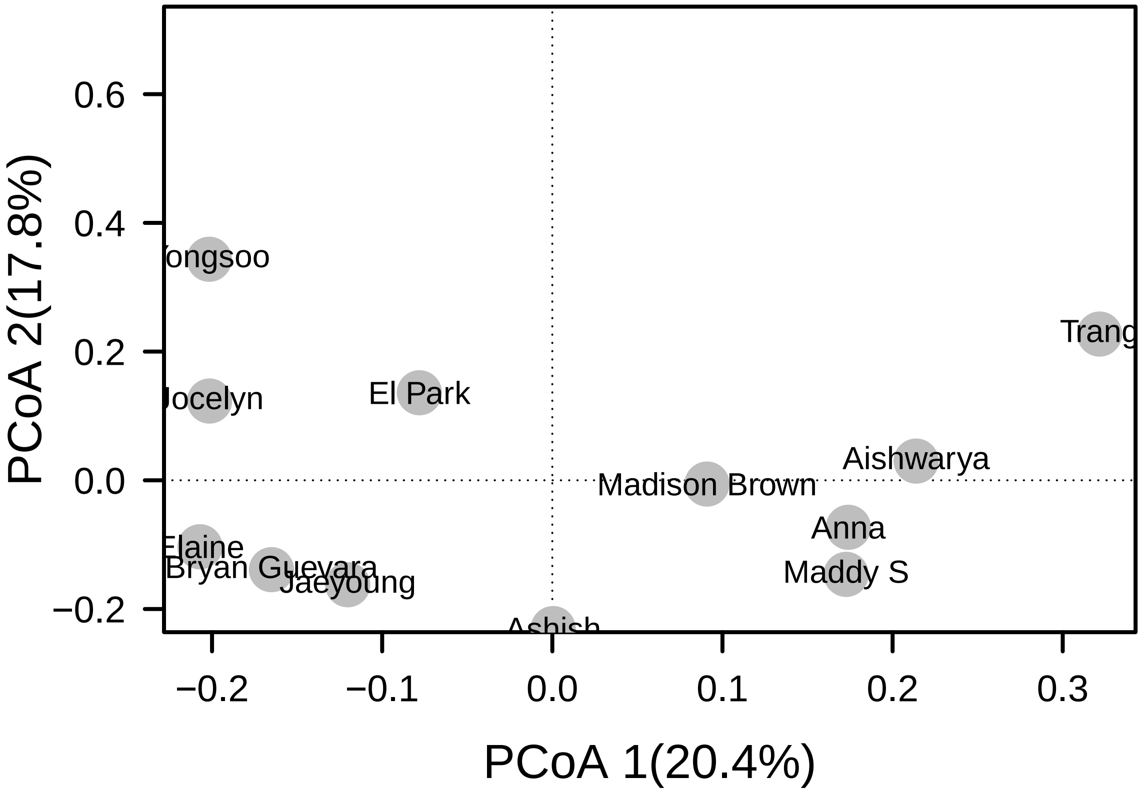


> ***Answer 1***: All the points on the curve are below the ab line, meaning individual efforts are not capturing the true diversity of the community, meaning collective sampling efforts are needed.

Question 2: Starting with the site-by-species matrix, visualize beta diversity. In the code chunk below, conduct principal coordinates analyses (PCoA) using both an abundance- and incidence-based resemblance matrix. Plot the sample scores in species space using different colors, symbols, or labels. Which "species" are contributing the patterns in the ordinations? How does the choice of resemblance matrix affect your interpretation?

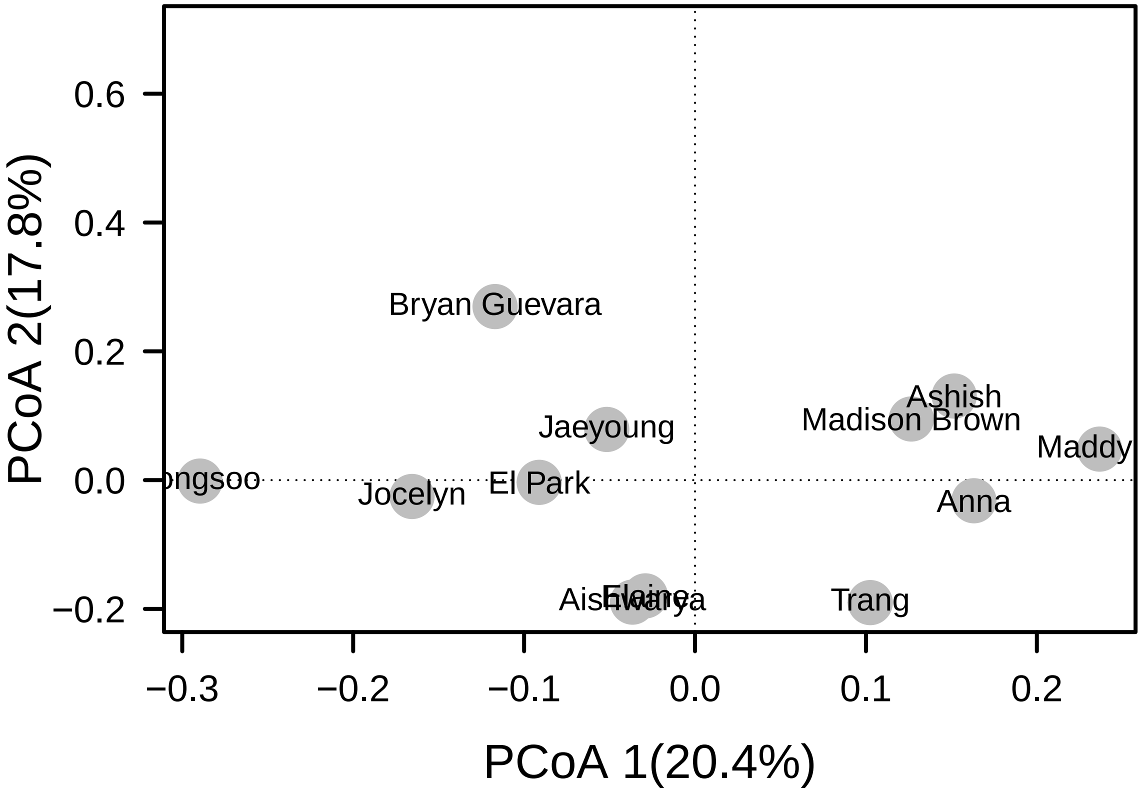
```r
jelly<-read.csv('./data/2JellyBelly_SbyS.csv', sep=",", header=TRUE, row.names=1)
#abundance
jelly.db<-vegdist(jelly, method="bray")
#pcoa abundance
jellyb.pcoa<-cmdscale(jelly.db, eig=TRUE, k=3)
explainvar1b<-round(jellyb.pcoa$eig[1]/sum(jellyb.pcoa$eig), 3)*100
explainvar2b<-round(jellyb.pcoa$eig[2]/sum(jellyb.pcoa$eig), 3)*100
explainvar3b<-round(jellyb.pcoa$eig[3]/sum(jellyb.pcoa$eig), 3)*100
sum.eigb<-sum(explainvar1b, explainvar2b, explainvar3b)
#plot
par(mar=c(5, 5, 1, 2)+0.1)
plot(jellyb.pcoa$points[,1], jellyb.pcoa$points[,2], ylim=c(-0.2, 0.7),
     xlab=paste("PCoA 1(",explainvar1b,"%)",sep=""),
     ylab=paste("PCoA 2(",explainvar2b,"%)",sep=""),
```

```r
      pch=16, cex=2.0, type="n", cex.lab=1.5,
      cex.axis=1.2, axes=FALSE)
axis(side=1, labels=T, lwd.ticks=2, cex.axis=1.2, las=1)
axis(side=2, labels=T, lwd.ticks=2, cex.axis=1.2, las=1)
abline(h=0, v=0, lty=3)
box(lwd=2)
points(jellyb.pcoa$points[,1], jellyb.pcoa$points[,2],
        pch=19, cex=3, bg="gray", col="gray")
text(jellyb.pcoa$points[,1], jellyb.pcoa$points[,2],
      labels=row.names(jellyb.pcoa$points))
```




```r
#incidence
jelly.ds<-vegdist(jelly, method="bray", binary=TRUE)
#pcoa incidence
jellys.pcoa<-cmdscale(jelly.ds, eig=TRUE, k=3)
explainvar1s<-round(jellys.pcoa$eig[1]/sum(jellys.pcoa$eig), 3)*100
explainvar2s<-round(jellys.pcoa$eig[2]/sum(jellys.pcoa$eig), 3)*100
explainvar3s<-round(jellys.pcoa$eig[3]/sum(jellys.pcoa$eig), 3)*100
sum.eigs<-sum(explainvar1s, explainvar2s, explainvar3s)
#plot
par(mar=c(5, 5, 1, 2)+0.1)
plot(jellys.pcoa$points[,1], jellys.pcoa$points[,2], ylim=c(-0.2, 0.7),
      xlab=paste("PCoA 1(",explainvar1b,"%)",sep=""),
      ylab=paste("PCoA 2(",explainvar2b,"%)",sep=""),
      pch=16, cex=2.0, type="n", cex.lab=1.5,
      cex.axis=1.2, axes=FALSE)
axis(side=1, labels=T, lwd.ticks=2, cex.axis=1.2, las=1)
axis(side=2, labels=T, lwd.ticks=2, cex.axis=1.2, las=1)
```

```r
abline(h=0, v=0, lty=3)
box(lwd=2)
points(jellys.pcoa$points[,1], jellys.pcoa$points[,2],
       pch=19, cex=3, bg="gray", col="gray")
text(jellys.pcoa$points[,1], jellys.pcoa$points[,2],
     labels=row.names(jellys.pcoa$points))
```



Answer 2: When considering abundance, we lessen the weight of rare jellies, and when we consider incidence, we increase the weight of rare jellies. I think it's interesting that in the abundance pcoa, my jelly data is very isolated from others, and in the incidence based pcoa, my data is similar to multiple others'. This may mean we have similar rare species but a dissimilar overal community composition.

Question 3 Using the preference-profile matrix, determine the most popular jelly bean in the class using a control structure (e.g., for loop, if statement, function, etc).

Answer 3:

```r
prefs<-read.csv('/cloud/project/QB2025_Park/Week5-Confection/data/JellyBelly_PreferenceMatrix.csv', sep
max_sum <- -Inf
most_popular_jelly_bean <- ""
for (i in 2:ncol(jelly)) { # Start from 2 to skip the first column (names)
  column_sum <- sum(jelly[, i], na.rm=TRUE)

  # Step 4: Use an if statement to update the most popular jelly bean
  if (column_sum > max_sum) {
    max_sum <- column_sum
    most_popular_jelly_bean <- colnames(jelly)[i]
  }
}
```

```r
cat("The most popular jelly bean is:", most_popular_jelly_bean, "with a total preference score of", max
```

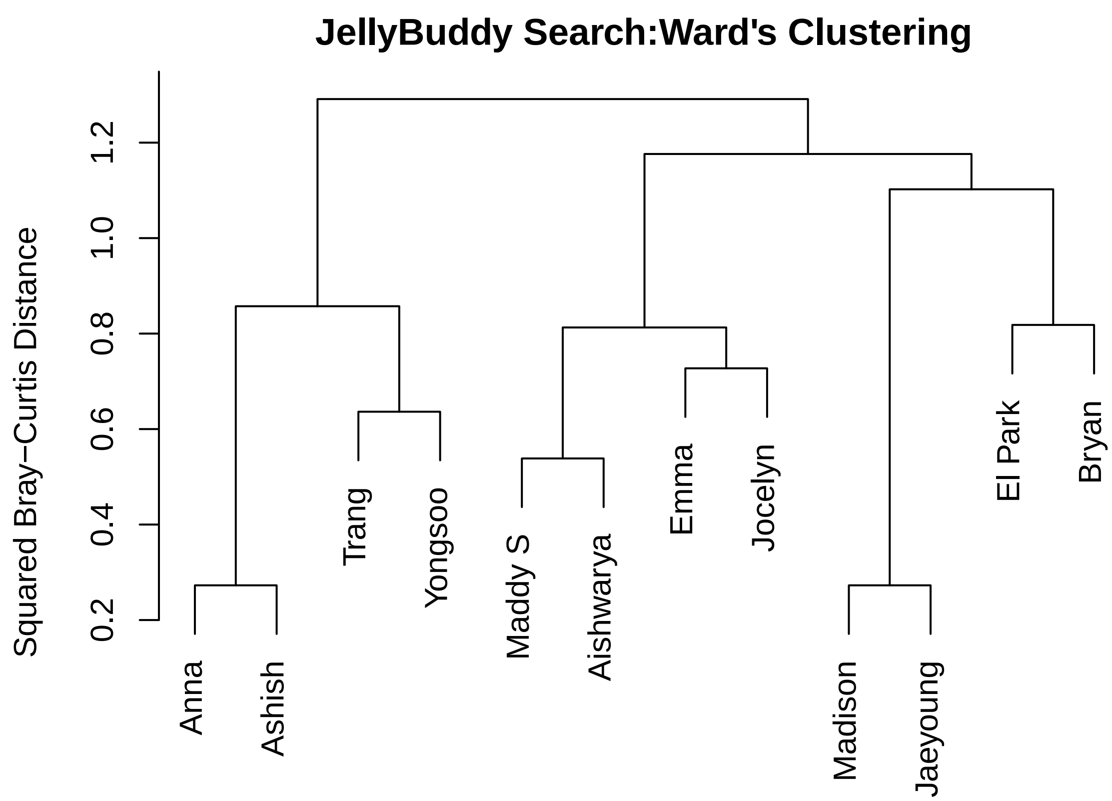
```
## The most popular jelly bean is: Raspberry with a total preference score of 67 !
```

Question 4 In the code chunk below, identify the student in QB who has a preference-profile that is most like yours. Quantitatively, how similar are you to your "jelly buddy"? Visualize the preference profiles of the class by creating a cluster dendogram. Label each terminal node (a.k.a., tip or "leaf") with the student's name or initials. Make some observations about the preference-profiles of the class.

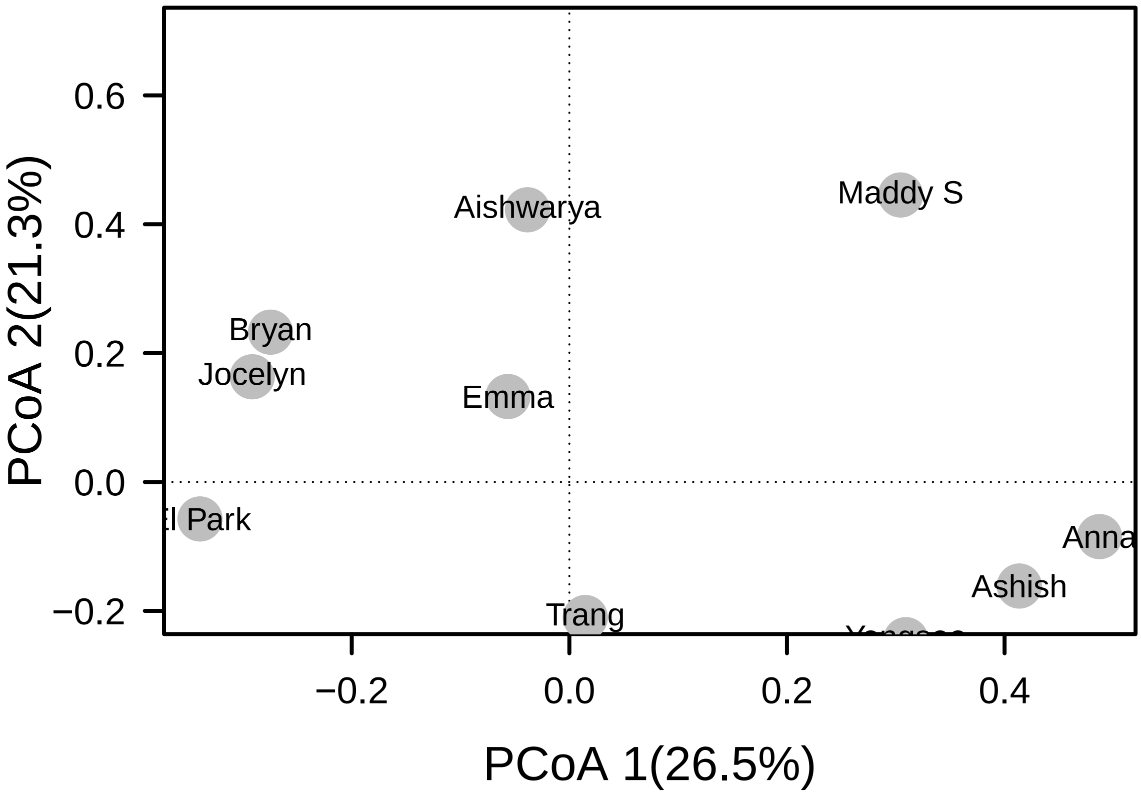
```r
require('viridis')
prefs<-na.omit(prefs)
pref.db<-vegdist(prefs, method= "bray")
print(pref.db)
```

```
##             El Park     Trang   Madison      Emma   Maddy S      Anna  Jaeyoung
## Trang     0.8333333
## Madison   0.8333333 0.8000000
## Emma      0.8947368 0.6470588 0.7647059
## Maddy S   1.0000000 1.0000000 1.0000000 0.7647059
## Anna      1.0000000 0.7777778 1.0000000 0.7500000 0.7777778
## Jaeyoung  0.6923077 0.8181818 0.2727273 0.6666667 1.0000000 0.8000000
## Jocelyn   0.8823529 0.8666667 0.7333333 0.7272727 0.8666667 1.0000000 0.7500000
## Bryan     0.8181818 1.0000000 1.0000000 0.8750000 1.0000000 1.0000000 1.0000000
## Aishwarya 0.8666667 0.8461538 0.8461538 0.6000000 0.5384615 0.8333333 0.8571429
## Yongsoo   1.0000000 0.6363636 0.8181818 0.8888889 0.8181818 0.6000000 0.8333333
## Ashish    0.8571429 0.8333333 0.8333333 0.7894737 0.6666667 0.2727273 0.6923077
##             Jocelyn     Bryan Aishwarya   Yongsoo
## Trang
## Madison
## Emma
## Maddy S
## Anna
## Jaeyoung
## Jocelyn
## Bryan     0.8571429
## Aishwarya 0.6666667 1.0000000
## Yongsoo   0.7500000 1.0000000 1.0000000
## Ashish    1.0000000 1.0000000 0.8666667 0.5384615
```

```r
pref.w<-hclust(pref.db, method="ward.D2")
par(mar=c(1, 5, 2, 2)+0.1)
plot.new()
plot(pref.w, main="JellyBuddy Search:Ward's Clustering",
     ylab="Squared Bray-Curtis Distance")
```




```r
prefd.pcoa<-cmdscale(pref.db, eig=TRUE, k=3)
explainvar1b<-round(prefd.pcoa$eig[1]/sum(prefd.pcoa$eig), 3)*100
explainvar2b<-round(prefd.pcoa$eig[2]/sum(prefd.pcoa$eig), 3)*100
explainvar3b<-round(prefd.pcoa$eig[3]/sum(prefd.pcoa$eig), 3)*100
sum.eigb<-sum(explainvar1b, explainvar2b, explainvar3b)
#plot
par(mar=c(5, 5, 1, 2)+0.1)
plot(prefd.pcoa$points[,1], prefd.pcoa$points[,2], ylim=c(-0.2, 0.7),
     xlab=paste("PCoA 1(",explainvar1b,"%)",sep=""),
     ylab=paste("PCoA 2(",explainvar2b,"%)",sep=""),
     pch=16, cex=2.0, type="n", cex.lab=1.5,
     cex.axis=1.2, axes=FALSE)
axis(side=1, labels=T, lwd.ticks=2, cex.axis=1.2, las=1)
axis(side=2, labels=T, lwd.ticks=2, cex.axis=1.2, las=1)
abline(h=0, v=0, lty=3)
box(lwd=2)
points(prefd.pcoa$points[,1], prefd.pcoa$points[,2],
       pch=19, cex=3, bg="gray", col="gray")
text(prefd.pcoa$points[,1], prefd.pcoa$points[,2],
     labels=row.names(prefd.pcoa$points))
```



Answer 4: Jaeyoung has the least dissimilarity to me, but the difference is still 0.69 out of 1, so I don't seem to have very similar tastes to anyone else. I find it interesting that though we have the least dissimilarity, I am clustered with Brian. Additionally, PCOA shows that Jocelyn is actually closer to me than Brian.

SUBMITTING YOUR ASSIGNMENT

Use Knitr to create a PDF of your completed `7.DiversitySynthesis_Worksheet.Rmd` document, push it to GitHub, and create a pull request. Please make sure your updated repo includes both the pdf and RMarkdown files.

Unless otherwise noted, this assignment is due on **Wednesday, February 19$^{\mathbf{th}}$, 2025 at 12:00 PM (noon)**.